SpecTool — CSI Spec Generator

Release status: SpecTool 1.0 — part of AVTools Suite Pro v3.1.0, for Revit 2025, 2026 and 2027. Additional bundled section starters are on the roadmap. Report issues to support@bimacoustics.net.

New to SpecTool? Start with the *SpecTool Quick Start* (2 pages). This guide is the full reference.

0) What SpecTool Is

A Revit add-in wizard that reads the AV equipment in the current project, generates CSI MasterFormat **Part 2 (Products)** articles via Claude matched to the consultant's own house format, fills gaps from datasheet PDFs, and **splices the result into a per-section boilerplate .docx** — with an audit-friendly Create-or-Update flow.

- **Audience:** AV consultants, integrators and spec writers using Revit.
- **Scope:** Divisions **11** (equipment) and **27** (communications / AV).

Registry-extensible for others.

- **Edition:** Pro-only feature. Standard users see the ribbon button but

get a paywall prompt.

- **Ribbon location: BIM Acoustics → AV Tools → SpecTool**. Hover over the

button and press F1 to open its help page on bimacoustics.net.

- **Internet connection:** required for the AI steps, which run through

the BIM Acoustics cloud service: **Classify unknowns (AI)**, **Find datasheets on the web (AI)**, From URL... **downloads, and** Generate with AI**. The model scan, Setup, and datasheet-folder matching work offline, but a spec can't be generated without a connection.

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1) Prerequisites — Model + Family State Drive Spec Quality

SpecTool does not invent product data. Every value that lands in the generated spec traces back to one of three sources: a **Revit family Type parameter**, a** placed instance count**, or a** PDF datasheet**. Garbage in, garbage out — but very fixable garbage, once the input sources are named.

1.1 Family loading — what the scan sees

The AV Products scan reads only from a whitelist of Revit categories: Audio Visual Devices, Communication Devices, Data Devices, Speciality Equipment, Generic Model, Electrical Fixtures, Electrical

Equipment.

- **Loaded but never placed** → filtered out. A family Type must have at

least one placed instance to appear as a row on Products step 2.

- **Wrong category** → invisible to the scan. Use **+ Add from model** on

step 2 to pull it in anyway; the tool won't guess at categories the scanner is conservative about.

- **Model parameter blank** → filtered out (row appears in the Skipped

banner, folded up). No Model = no identity to spec against.

Fix once, iterate forever: get the family into the right category and give every Type a Model value before running the wizard. The scan result is the entire universe SpecTool works with.

1.2 Type parameters — the spec's raw material

Each Product row's technical values are read from Type-level parameters, in this order of importance:

| Parameter | What it drives | Missing → | |---|---|---| | **Model** | Row identity; datasheet folder auto-match; classification input | Row skipped | | **Manufacturer** | "Basis of Design: <mfg> <model>"; datasheet URL disambiguation | Blank BoD line; datasheet auto-match may still hit | | **Description** | Fills SYSTEM DESCRIPTION prose; drives AI generation when no datasheet | Generic "audio-video component" language | | **Datasheet URL** | Family-URL auto-download tier | Skip that tier; fall through to folder / manual / AI hunt | | **Type/Sub codes** | AVIXA drawing type identity; appears as column and PR2 label context | Blank column; less specific spec text |

Live parameter-count check on Setup step 1: the model-parameters section shows a live count of family Types carrying a value for each picked parameter. A count of 0 means either no Types have that value or the wrong parameter name is picked — the count is the single fastest signal that model data is thin.

1.3 Standard vs custom parameter names

SpecTool ships knowing the BIM Acoustics / JSBA family parameter names by default. For firms whose families use different names (e.g. `MFG` instead of `Manufacturer`, `Cat #` instead of `Model`):

- **Setup step 1 → Model parameters** — six pickers remap SpecTool's

logical roles onto the actual family parameter names in the current project.

- **Re-run the parameter-count check after remapping.** A picker with 0

values still means empty data — remapping only helps if the values exist somewhere.

1.4 Type + family naming — classification input

The keyword classifier reads family and Type names to place products in the right Part 2 article. Clear names classify without help:

- `"JSBA Loudspeaker CF10 Full-Range Ceiling"` → **LOUDSPEAKERS ✓**

- `"AV Amplifier – Crown DCi 4x300"` → **POWER AMPLIFIERS ✓**
- `"Speaker3"` → **unclassified** (needs manual assign on step 2 or 4)
- `"AV Device"` → **unclassified** (too generic)

Manual overrides on Products step 2 or Categories & Articles step 4 always win, and category corrections are learned across sessions — teaching once is permanent.

1.5 Placement quantity — the QTY column

The **Quantity** in the generated spec is a live count of placed FamilyInstances of each Type. Not from a Type parameter, not from a schedule — the model itself is the count of record.

- **Un-placed Types** don't appear on the spec even if the family is

loaded. Place at least one instance to spec.

- **Same model across two families** (e.g. a Cohesion CO10 loaded twice

under different family templates) is merged on the Model string during classification. The spec sees one row, quantity summed.

1.6 Type-level shared parameters (JSBA_ family) — extraction targets

The registry lists **key_specs** per category — the technical values a spec cares about for that product class (impedance, wattage, dispersion, mount type, etc.). SpecTool looks for those values in this order:

1. **JSBA_ shared parameters on the Type** — if the family has been stamped with `JSBA_Impedance_Ohms = 8`, that lands in the spec directly. No datasheet needed. 2. **PDF datasheet extraction** — if the shared parameter is empty, the datasheet processor reads the PDF for the field. 3. **AI generation from family Description** — last resort; often accurate but always worth reviewing. 4. **"as scheduled on the Drawings"** — printed when all three above fail. The spec still issues; the value is deferred to the Drawings.

Payoff for stamping shared parameters at family-authoring time: every future spec for that product is deterministic and doesn't need the datasheet PDF present. **This is the highest-leverage place to invest time** if a firm plans to reuse families across projects.

1.7 Boilerplate .docx state

The **boilerplate** Word document supplies Part 1 (General) and Part 3 (Execution). Its state matters:

- **Styles must exist and be named** as the template expects (PRT, ART,

PR1, PR2, PR3). A boilerplate that uses `Normal` throughout still splices, but the output doesn't paginate or number cleanly.

- **Placeholder text** (`ARTICLE` , `SUBPART` , `PARAGRAPH`) is scaffolding

— the tool suppresses these from stray warnings, but they'll appear in the output verbatim. Delete them by hand or replace with real content.

- **No boilerplate on hand?** The Use bundled starter button on Setup

step 1 loads UFGS 27 41 00 (Div 27 audio-video) as a drop-in — see §10.

1.8 Template .docx state

The **project spec template** (architect-provided) supplies styles, fonts, headers/footers, page numbering. Its state matters for the OUTPUT format, not for content:

- Missing style definitions → splice writes raw text without style

application. Cure by importing the missing styles into the template.

- Header/footer/numbering baked into the template is preserved on

Create — the output looks like everything else in the project book.

1.9 Datasheet folder organization

The auto-match tier searches filenames for the Model string, recursively. The lighter-touch conventions that make it hit reliably:

- **Filename contains the model number** (with or without dashes,

spaces, underscores). `Cohesion_CO10.pdf`, `CO10 Data Sheet.pdf`, `co10-spec.pdf` all match model `CO10`.

- **Manufacturer subfolders help but aren't required.** The recursion

handles depth; the auto-match only looks at filenames.

- **Corrupt or non-text PDFs** (scanned images without OCR) won't

extract values. The datasheet auto-matches, but PDF extraction falls back to AI-only generation from the description.

1.10 The iteration rule — fix the model, not the doc

Because Update mode re-splices tool-owned Part 2 articles on every issue, **hand-editing a Part 2 article in the .docx is temporary**. The next issue will overwrite it.

- **Wrong Manufacturer / Model?** Fix it on the family Type in Revit,

then re-run.

- **Missing performance value?** Stamp the shared parameter on the Type,

or place the datasheet PDF in the folder — then re-run.

- **Wrong article assignment?** Fix on step 2 or step 4 — the correction

is learned.

- **Wrong Part 1 or Part 3 content?** Hand-edit the .docx — those are

outside the tool's replacement zone and survive Update.

- **Want a phase's spec frozen?** Use Start Fresh on Review step 5 to

fork a new record — future edits go to the new file, the old one stays untouched.

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2) The Wizard at a Glance

Five ordered steps plus a section-checklist **Home**:

Home (Sections) → 1 Setup → 2 Products → 3 Data & Datasheets → 4 Categories & Articles → 5 Review & Generate

- **Non-destructive navigation:** moving between steps never re-runs the

Revit scan, never re-calls the AI, and never loses ticks, edits, or datasheet attachments. Those fire only on explicit buttons.

- **One scan per project:** the AV Products scan runs once and the cached

inventory serves every section via "Start Another Section."

- **Section-scoped:** the wizard runs one CSI section at a time. Multi-

section projects are handled by returning to Home between runs.

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3) Section Checklist (Home)

The landing screen. Lists every CSI spec section the current project needs, based on the equipment scan and any prior issues on disk. The first time the page opens in a session (with a project open) it runs the AV scan automatically, so every section is listed — and mergeable — up front.

Columns: Section number · Title · Products count · Articles count · Status.

Row buttons:

- **Work on it** → loads the section into Setup and jumps to step 1.
- **Remove** forgets one specific issue of that section from the project

record (the output .docx on disk is **never** deleted; only the tool's pointer to it is dropped). Multiple issues of the same section (Minimal + BoD, 35% CD + 100% CD) each get their own row and can be removed independently.

- **Merge…** folds one section into another so a one-off item doesn't

force a separate spec. Example: a lone patchbay classifies into `27 15 00` (Communications Cabling); merge `27 15 00 → 27 41 00` and the patchbay keeps its own article but generates inside 27 41 00. The merged row shows "→ Merged into 27 41 00", the target shows "incl. 27 15 00", and **Work on it** on a

merged row opens the target. Category labels on the Products/Categories pages show the redirect, e.g. " (27 15 00 → 27 41 00)". Merges are saved per project and reversible from the same dialog.

Header buttons:

- **Scan AV Products** — same event as step 2's Scan; refreshes the checklist.
- **New Section** — start a section the scan didn't surface (any division

and number the user types).

- **Refresh** — re-read persisted state after external changes.

Row rules:

- One row per persisted issue (per output file), latest first by generation date.
- Sections with products in the scan but no issued output show as

"Pending — nothing issued yet."

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4) Setup (step 1)

Section identity and the documents that drive generation. Every field is remembered.

Section

- **Division** — dropdown from the registry: 11 and 27. (The list also

shows 28, electronic safety and security, which is a preview.)

- **Section** — combo picker per division; **editable**, so a section not in

the standard list (custom firm sections, or a section number the picker hasn't been updated for) can be typed directly, including title. Example: `27 41 13 – AV for Auditoriums`.

- **Phase** — freeform, e.g. `35% CD`, `100% DD Rev 1`. Written into the

output filename and into the document header/footer (see §8).

- **Issue date** — defaults to today; remembered per project. Written into

the document header/footer on Create / Update.

Detail level

Four modes — see §9 for the deep dive.

User Boilerplate Detail (default; match your exemplar) Basis of Design (BoD + "Substitutions: per Section 01 25 00.") Minimal Spec (leanest) All Mfg Specs (every stated value as a bullet)

Filename (optional override)

Leave blank → auto-name as `<section> – <title>_<phase>.docx` in the Output folder. Type anything → that string is used verbatim; `.docx` is appended if missing.

Documents & folders

- **Project spec template (format)** — the architect's spec .docx, supplies

the FORMAT: styles, fonts, header/footer, numbering.

- **Your boilerplate (content)** — the firm's spec .docx, supplies Part 1

(General) and Part 3 (Execution) content.

- **Datasheet folder** — manufacturer datasheet library (PDFs plus any

saved product pages), searched recursively.

- **Output folder** — where the generated .docx lands.

"Use bundled starter..." button

Appears when a **UFGS starter** exists for the current section (today: only `27 41 00`). Loads the verbatim NAVFAC UFGS 27 41 00 Part 1/3 (February 2026) plus a short AVIXA supplement (V201.01, V202.01, F202.01, RP-C303.01). The button hides for sections without a bundle.

Model parameters (advanced)

For firms that keep product data in non-standard Revit parameter names. Six pickers: Drawing Type code, Sub designation, Manufacturer, Model, Datasheet URL, Description. Live count of parameters carrying values across scanned AV family types — a wrong pick reads 0.

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5) Products (step 2)

The AV equipment inventory. Grid with one row per family+type instance found in the model, with an editable Category column.

Header buttons:

- **Scan AV Products** — runs the family scanner (main scan or rescan).

Rescan preserves per-model exclusions (un-ticked products stay un-ticked).

- **Classify unknowns (AI)** — one batched Haiku call over all rows where

the layered classifier couldn't decide. AI results are shown but **not auto-learned** — only user corrections are remembered.

- **+ Add manually...** — hand-add a product not in the model (software,

license, service). Dialog captures Model, Manufacturer, Type, Quantity, Description. Description is critical for AI-generated content since no datasheet exists.

- **+ Add from model** — pick families the scan filtered out (families

without a Model parameter, families in categories the scan is conservative about). Full checklist of every loaded family type. The dialog's **Category** combo at the top lets the user assign a category to the whole batch — options are labeled `<Article> (<Section>)` so the target section is visible before picking. Default is **Classify automatically**; choosing an explicit category **overrides the classifier** and locks the category (CategorySource = User) so subsequent auto-passes won't touch it.

Cross-section override (using Add from model to fold IT gear into an AV spec, or similar): On a 27 41 00 workflow, a consultant who doesn't want a separate 27 21 00 data-comm spec can click + Add from model, select a switch family, and set the picker's Category combo to any 27 41 00 article (MISCELLANEOUS, EQUIPMENT RACKS, etc. — whatever fits the article structure they want). The switch then lands in the 27 41 00 workflow instead of being routed to 27 21 00 by the classifier. Same mechanism works for keeping projection screens or projectors in an AV spec instead of separate Div 11 specs, or any other section split the project doesn't need. The label on the picker's category options (`<Article> (<Section>)`) is deliberate — it's the tell for whether a choice keeps the product in the current workflow's section or sends it elsewhere.

Inline "+ New..." in the Add-manually dialog: create a new registry category without leaving the Add Product flow.

Row columns: Include · Category (editable) · TypeCode · Sub · Manufacturer · Model · Notes.

Un-ticking a row **excludes** the product from the section. Exclusions survive rescans (keyed by Model, not by object reference).

Skipped banner — folded up until clicked. Shows families the scanner dropped (no Model value) so a wrong-scoped product isn't invisible.

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6) Data & Datasheets (step 3)

Where each included product's datasheet PDF gets associated. Only rows missing category-required key fields appear here.

The four-tier ladder (locked order)

1. **Datasheet folder auto-match** — filenames containing the model, searched recursively under the datasheet folder. **This wins over the session cache**: if a folder PDF appears, it overrides any prior URL-downloaded copy for the same model. Row status labels the override: `"Folder match (overrode <old>.pdf): <new>.pdf"` . When several files contain the model name, an **exact filename** (`XT850i.pdf`) beats a partial match (`TYPE LS_XT850i.pdf`), and a downloaded file beats a cloud-only placeholder. 2. **Family URL parameter** — auto-downloaded in the background when the step opens. Follows a product-

page link one level to find the first PDF link. 3. **From URL dialog** — manual URL entry, with a "Search web…" button that launches the user's default browser pre-searched for the device.

No PDF, only a product page? Many manufacturers (video displays especially) publish full specs on a web page with no downloadable datasheet. When a Family URL or From URL link yields a page with no usable PDF, SpecTool saves the page as `{model}.html` in the library and reads its specs at generation, just like a PDF. The row status reads "Saved product page" instead of "Downloaded". Limitation: only the page's static HTML is read — a spec table drawn by JavaScript after the page loads won't be seen.

Datasheets are read at Generate. A datasheet sitting in the library folder is read when you generate, even if you never opened this step. Correcting or replacing a datasheet and regenerating replaces the specs filled from the old one (values from the Revit family still win). 4. **Attach** — file-picker fallback. A copy is saved into the datasheet folder, filed by manufacturer subfolder, named by model, so it auto-matches forever after.

AI web hunt

Header button **Find datasheets on the web (AI)** — for every row still missing a datasheet, does a server-side Anthropic web_search + PDF follow, downloads the result, files it in the library. Use when the folder + URL tiers came up dry.

The hunt is deliberately strict — a wrong datasheet is worse than none:

- **PDF datasheets only.** It never files a web page. If it finds no PDF

the row reads "No PDF datasheet found on the web — use From URL or Attach" (From URL can still save a product page, above).

- **Single-product datasheets only.** User/installation manuals,

quick-starts, comparison charts, multi-product catalogs, price lists, reseller listings, case studies and warranty/return policies are rejected.

- **Content check.** A downloaded file is kept only if its own text names

the model (so a datasheet hosted on a parent company's site passes). A rejected file is deleted and the row says why. Very short model codes (under 3 characters) can't be verified and skip this check.

Clear button (per row)

Escape hatch: drops the session association for that row so every tier re-runs clean. Useful when a corrected model number is stuck to the prior model's PDF.

Anything still missing

Writes "as scheduled on the Drawings" in the generated spec — the row is still spec'd, just deferred to the Drawings for values.

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7) Categories & Articles (step 4)

Deliberate review moment. Same editable Category column as step 2, but grouped by **section → article** with unclassified rows on top. A consultant sees exactly which Part 2 article each product will land in and can correct assignments before generation.

- **+ Add category** — opens the Add Category dialog to create a new

registry entry. Reserved Part 1/3 article names (SUMMARY, SYSTEM DESCRIPTION, EXECUTION, INSTALLATION, and ~30 others per CSI SectionFormat) are blocked here — they'd collide with the boilerplate and get spliced into Part 2 by mistake.

- **Reload categories** — re-reads spec_categories.json without

restarting Revit. Use after hand-editing the JSON.

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8) Review & Generate (step 5)

The generation + splice step.

Mode indicator

Header shows either:

- **CREATE (splices into the project template)** — no prior issue for

this section number.

- **UPDATE (splices into the last issued doc)** — a prior issue exists,

and its output file is on disk. Update mode splices into the last-issued file so hand edits outside tool-owned articles survive.

- **CREATE — Start fresh (ignoring last issue)** — Start Fresh checkbox

is ticked; the wizard forces CREATE mode against the template.

Start fresh (ignore last issue)

Checkbox — only visible when a prior issue exists. Forces CREATE against the template instead of UPDATE against the last issued doc. Use when the prior output has stray articles, was generated from an obsolete registry, or represents a genuinely new engagement rather than the next phase of the same one.

Generate with AI (preview)

- One Sonnet call **per Part 2 article** (not per Type). A section with 10

articles = 10 API calls.

- Devices are grouped by category, articles ordered by **registry order**

(loudspeakers → power amplifiers → signal processing → ...), not alphabetical.

- **SYSTEM DESCRIPTION** is generated after the Part 2 articles and prepended

to the preview under `Article = "SYSTEM DESCRIPTION"`. 90-second hard timeout on that call so an Anthropic hang can't lock the wizard.

- **Auto-fallback to per-Type**: if a Part 2 article's response hits

`stop_reason=max_tokens` (common with All Mfg Specs + 15+ Types), the wizard automatically retries that article one Type at a time and merges the results. Status line reports the retry.

- **529 / 5xx retry**: transient Anthropic overload responses retry with

exponential backoff (2s → 5s → 12s → 30s, 5 attempts). Non-transient failures (400 / 401 / bad prompt) fail fast.

Preview grid

Every generated paragraph is a row with editable Style + Text and an **Inc.** checkbox at the front. Un-tick to drop a paragraph from the splice without losing it — re-tick to bring it back.

- **Single-click** toggles the checkbox (no double-click).
- **Shift-click a range of rows**, then click any checkbox in the range,

to flip every selected row to that value in one pass.

- **Delete key** removes a row entirely.
- **Style + Text** are editable; the splice writes what's in the grid.

Create / Update spec

Runs the pipeline:

1. Copy source → output (source = template on CREATE, last-issued doc on UPDATE). 2. Lift Part 1 (GENERAL) from boilerplate into output. 3. Lift Part 3 (EXECUTION) from boilerplate into output. 4. Splice SYSTEM DESCRIPTION into Part 1 (after SUMMARY). 5. Splice each Part 2 article (highlighted yellow for review). 6. Update Section Includes (Part 1 SUMMARY) with the current section's articles. 7. Post-splice **reorder pass**: Part 2 articles get shuffled into registry order (unknown articles preserved at end). 8. Report strays: articles in Part 2 not in the current run and not in the registry get flagged in the status line for user review. 9. **Refresh header/footer fields** — section number, section title, phase, and issue date. The template's placeholders ("SECTION TITLE", "00 00 00", the prior phase and date) are replaced with this issue's values in the template's own format: a "100% Design Development" header becomes "50% Construction Documents", "100% DD" becomes "50% CD", and a day-first date stays day-first. Explicit tokens `{{SECTION_NUMBER}}`, `{{SECTION_TITLE}}`, `{{PHASE}}`, `{{DATE}}` also work. A field with no recognizable placeholder is left alone.

After a successful run

- **Open document** launches the .docx in Word.
- **Start another section** returns to Home with the scan + folders + phase +

template retained; boilerplate is cleared (per-section content).

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9) Detail Levels (Deep Dive)

Four modes drive how rich the generated Part 2 articles are. Structure (Type label → description → Acceptable Products → substitutions) is the same across all four; density and framing differ.

User Boilerplate Detail (default)

Match the consultant's own boilerplate article EXACTLY — depth, voice, which fields get bullets. Recommended default. Falls back gracefully when spec_params are sparse.

Basis of Design

CSI "open-proprietary" convention:

- Type block opens with `Basis of Design: <manufacturer> <model>` as the

first PR3 line under Acceptable Products.

- Every registered key spec is a PR3 bullet — absent values read

`as scheduled on the Drawings`.

- Type block closes with `Substitutions: per Section 01 25 00.`

Use when the drawings/schedule govern absolute values and the spec is the enforceability layer.

Minimal Spec (leanest)

Absolutely minimal — Type label + ONE PR2 description sentence + Acceptable Products + product line(s) + Accepted Substitutions. **Nothing else.** No `Key performance:` label. No PR3 performance bullets even when spec_params carries values. Performance details live on the Drawings and datasheet at this level.

Use for quick specs, back-of-house scope, or when the model is the authoritative document.

All Mfg Specs

Every performance value present in spec_params (plus web-hunted fields when a datasheet was extracted) becomes its own PR3 bullet, grouped logically. `Key performance:` label is **omitted** — every spec at this level IS a performance value, so the label is redundant.

Use for federal/spec-heavy work where the spec is the primary enforceability document and drawings are supplemental.

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10) Bundled Starters (UFGS)

What's bundled

UFGS 27 41 00 Audio-Visual Systems (February 2026, NAVFAC) — verbatim SpecsIntact SEC XML embedded in the tool. Part 1 (11 articles) and Part 3 (15 articles) parse into CSI-styled paragraphs (PRT/ART/PR1/PR2/PR3) that splice directly — no docx round-trip.

AVIXA supplement (BIM Acoustics-curated) merges four additional standards into the UFGS REFERENCES article:

- ANSI/AVIXA V201.01 (2021) Image System Contrast Ratio
- ANSI/AVIXA V202.01 (2016) Display Image Size for 2D Content (DISCAS)
- ANSI/AVIXA F202.01 (2020) Sightline Design for Video Displays
- ANSI/AVIXA RP-C303.01 (2018) Security in Networked AV Systems

How it's offered

Setup step 1 shows a **Use bundled starter...** button when the current section has a bundle. Clicking loads it as if it were a boilerplate file path (using an internal sentinel).

The Create/Update pre-flight dialog on a section with a bundle offers three choices:

- **Yes** — use the bundled starter.
- **No** — go back to Setup and pick your own boilerplate.
- **Cancel** — continue with Part 2 only (skeleton Part 1/3 from the template).

What's NOT bundled (yet)

- **Div 11** (screens, projectors, lifts) — UFGS does not publish equipment-

category starters. The "no boilerplate" dialog says so explicitly. Firms use their own AE boilerplate for equipment.

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11) Sections and Categories

The registry (`spec_categories.json` , embedded + disk-override) drives:

- Which sections show in the Setup picker per division.
- Which category options show in the editable Category dropdowns.
- Which section each category maps to (`section: null` = default 27 41 00).
- Which fields the datasheet extractor asks for per category.
- Which keywords classify a scanned family into a category.

Sections offered in the picker

Division 11: 11 52 00 (Audio-Visual Equipment / lifts), 11 52 13 (Projection Screens), 11 52 16 (Projectors).

Division 27: 27 15 00 (Communications Horizontal Cabling), 27 21 00 (Data Communications Network Equipment), 27 41 00 (Audio-Video Systems), 27 41 16 (Integrated AV), 27 51 13 (Paging), 27 51 19 (Sound Masking), 27 51 23 (Intercom).

Categories in the registry

Div 27 AV (default → 27 41 00): LOUDSPEAKERS, POWER AMPLIFIERS, SIGNAL PROCESSING AND CONTROL, WALL PLATES (OPERATIONS PANELS), VIDEO DISPLAYS, SOFTWARE AND LICENSES, COMPUTERS, EQUIPMENT RACKS, UNINTERRUPTIBLE POWER SUPPLY, SOURCE EQUIPMENT, MICROPHONES, MIXERS, SWITCHERS, DISPLAY MOUNTS.

Div 27 Data Communications (IT / network scope):

- **NETWORK EQUIPMENT** → 27 21 00 — network switches (managed, PoE,

AV/data), routers, firewalls, wireless access points, media converters, PoE injectors, SFP modules.

- **COMMUNICATIONS CABLING** → 27 15 00 — patch panels/bays, patch cords,

faceplates, keystones, wall jacks, structured cabling (Cat5e / 6 / 6A / 7 / 8, fiber), plenum/riser bulk cable.

Div 11 (motorized AV equipment):

- **PROJECTION SCREENS** → 11 52 13 — fixed, tab-tensioned, and motorized

screens (`"motorized screen"` in the classifier).

- **PROJECTORS** → 11 52 16 — laser, DLP, LCD, short-throw, UST.
- **LIFTS** → 11 52 00 — motorized projector/display/TV lifts, ceiling-

recessed and drop-down lifts, motorized flip/concealment mounts. The longer-keyword-wins rule ensures `"Projector Lift"` families classify as LIFTS (not PROJECTORS).

Catch-all: MISCELLANEOUS — user-assigned only; no keywords match, so nothing auto-classifies here.

Adding a custom category

Add Category dialog on step 4 (or the "+ New..." inline button on Add manually in step 2) opens a form: Article name, key, section, keywords, optional key specs. Writes to spec_categories.json.

Guardrail: article names that collide with the CSI Part 1 or Part 3 catalog (SUMMARY, REFERENCES, SUBMITTALS, EXECUTION, INSTALLATION, ~30 others) are rejected with a clear error — those belong in the boilerplate, not in Part 2.

Editing the registry directly

`spec_categories.json` next to `JSBA.SpecGen.dll` (disk copy wins over the embedded fallback). Reload without restarting Revit via the Reload Categories button on step 4.

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12) Update Mode Carries Edits Forward

When a section has a prior issued .docx, Create/Update runs in **UPDATE** mode by default:

- **Source** = the last issued .docx (not the project template).
- Part 1 and Part 3 are **relifted** from the current boilerplate (so a

boilerplate change propagates).

- Part 2 articles named in the current run are **replaced** in place.
- Part 2 articles NOT named in the current run are **preserved** (a hand-

written article between phases stays).

- SUMMARY's Section Includes list has any new phrases appended.

Consequence: hand edits inside tool-owned articles are OVERWRITTEN on Update; edits outside them survive. To preserve tool-owned-article edits across issues, either write them elsewhere or use Start Fresh to fork a new record.

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13) Filename Convention

Auto: `<section> – <title>_<phase>.docx` in the Output folder. Example: `27 41 00 – Audio-Video Systems_100%CD.docx`.

User override: the Filename field on Setup step 1. Value used verbatim in the Output folder; `.docx` appended if missing. Path-separator characters are stripped so the override can't escape the folder.

Per-section: the override is cleared by Start Another Section along with the section number and boilerplate, so a filename that fit 27 41 00 doesn't leak into the next section's output.

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14) Data Confidentiality (Tier 1)

The prompt sent to Anthropic carries ONLY generic content:

SENT:

- Section number and title (e.g. "27 41 00 - Audio-Video Systems").
- Article name (e.g. "POWER AMPLIFIERS").
- Category prompt guidance.
- Equipment list: manufacturer, model, quantity, Type/Sub codes,

spec_params, description, doc_url.

- Detail level.
- The consultant's boilerplate exemplar (their own article, style + text).

NOT SENT — asserted by test harness before every request:

- Project name.
- Building name.
- Owner name.
- Architect name.
- Consultant name.
- Any address or geographic identifier.

Diagnostic dumps of every prompt and response land in `%APPDATA%\BimAcoustics\SpecTool\` for audit — file name is the article (`specgen_prompt_LOUDSPEAKERS.txt` , `specgen_response_LOUDSPEAKERS.txt`).

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15) Troubleshooting

"Generation failed: 529 overloaded_error"

Anthropic upstream overload — transient. The chat client already auto-retries 4 times with exponential backoff (2s → 5s → 12s → 30s ≈ 50s total). If the error still surfaces, the overload lasted longer than the backoff. Wait a minute and click Generate again.

"Add from model failed: Object reference not set to an instance of an object"

Prior fixes covered the two known NRE paths (BeginInvoke, XAML load race). If it recurs, check `%APPDATA%\BimAcoustics\SpecTool\add-from-model_error.txt` — the wizard dumps the full stack trace on the next failure.

"SYSTEM DESCRIPTION generation timed out — skipping"

The generator has a hard 90-second timeout. When it fires, the article is dropped (Part 2 articles are unaffected) and `%APPDATA%\BimAcoustics\SpecTool\specgen_response_SYSTEM DESCRIPTION.txt` records the reason.

"Strays in Part 2 (not in registry, not in this run — review)"

Articles present in Part 2 that neither the current run generated nor the registry knows about. Common causes:

- Historical artifacts from an older registry version (article names

that got consolidated or renamed).

- Boilerplate-only articles that the current registry doesn't need

(WIRE & CABLE, FABRICATION — these are legitimate).

- Hand-written articles between phases.

The tool never auto-deletes strays. Delete manually in Word if unwanted; leave in place if legitimate.

CSI template placeholders (`ARTICLE` , `SUBPART` , `PARAGRAPH`) are suppressed from the stray warning — they're style scaffolding, not real content.

Section Includes shows unexpected +N items

The wizard's UpdateSectionIncludes finds the SUMMARY article INSIDE Part 1 GENERAL (not just anywhere the text "Section Includes" appears). If your boilerplate has an unusual Part 1 structure and the anchor lands wrong, the fallback finds the first "Section Includes" text and walks forward from there.

Add Category refuses a name

The article name matches a CSI Part 1 or Part 3 article (SUMMARY, SYSTEM DESCRIPTION, EXECUTION, INSTALLATION, and ~30 others). Those belong in the boilerplate, not in Part 2. Pick a product-specific name (WAYFINDING DISPLAYS, PAGING MICROPHONES, etc.).

Spec quality feels thin, generic, or wrong

Usually a model-state problem, not a tool problem — walk back through §1:

- Are Type-level Manufacturer / Model / Description populated? (§1.2)
- Setup step 1's parameter-count check reading > 0 across the pickers?

(§1.2 + §1.3)

- Are JSBA_ shared parameters on the Type carrying the key_specs the

registry wants? (§1.6)

- Are datasheet PDFs in the folder with the model number in the filename?

(§1.9)

- Are the boilerplate's paragraph styles named PRT/ART/PR1/PR2/PR3?

(§1.7)

Fix in the model → re-run. The spec is a derived artifact; editing the .docx directly gets overwritten on the next issue.

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16) Lucius Capability Surface — SpecTool

What Lucius CAN do

- **Explain the wizard's workflow** — what each step does, what each

button does, what happens on Create vs Update, what Start Fresh means, when to use which Detail Level.

- **Explain a status message** — what a "Strays in Part 2" warning means,

why a 529 happened, why SYSTEM DESCRIPTION timed out.

- **Explain the registry schema** — how a category maps to a section, how

key_specs drive the datasheet extractor, how keywords drive classification.

- **Explain the UFGS bundled starter** — what's in it, why Div 11 doesn't

have one, when to use "Use bundled starter" vs a firm boilerplate.

- **Diagnose "why is my spec thin?"** — walk the user back through the

§1 chain (family loading, Type parameters, shared parameters, datasheets, boilerplate styles) and identify which link is empty.

- **Explain cross-section overrides** — the "I don't want a separate

27 21 00 spec — how do I keep my network switches in the 27 41 00 AV spec?" question. Answer: **+ Add from model** on step 2, set the picker's Category combo to a 27 41 00 article (its section shows in the label), pick the switch families. Or edit the Category column on step 2 directly for any row already added. Same trick works for keeping projectors/screens in an AV spec instead of separate Div 11 specs.

- **Recommend a Detail Level** given a project type and use case.
- **Recommend a datasheet source** given the ladder tiers.
- **Interpret a diagnostic dump** if the user pastes contents of

`%APPDATA%\BimAcoustics\SpecTool\*.txt`.

What Lucius CANNOT do (no tools)

Lucius does NOT have tools for:

| Topic | What's missing | What Lucius should say | |---|---|---| | **Trigger the wizard** | No tool to open the wizard or advance a step | "Open SpecTool from the BIM Acoustics ribbon; I'll walk you through the step you need." | | **Read wizard session state** | No tool to read Session.AllDevices, Session.Runs, Session.BoilerplatePath, or any other in-memory wizard state | "I can't see what you have entered in the wizard right now — describe what you see and I'll advise." | | **Read family Type parameters** | No tool to inspect Manufacturer / Model / Description or JSBA_ shared parameters on Types | "I can't read your family Type parameters directly — open one in Family Types (or a schedule) and tell me what's in Manufacturer, Model, Description, and any JSBA_ shared parameters for the category." | | **Trigger Generate or Create/Update** | No tool to run generation or splice | "Click Generate with AI on step 5 to draft the articles, then Create / Update spec to write the .docx." | | **Read the generated .docx** | No tool to open or parse the output file | Direct user to open the .docx in Word; can explain expected structure. | | **Add / edit / remove a category** | No tool to modify spec_categories.json | Direct user to step 4's Add Category dialog; explain the fields; explain the reserved-name guardrail. | | **Attach / clear a datasheet** | No tool | Direct user to step 3; explain the ladder and the Clear button. | | **Change a Detail Level** | No tool | Direct user to Setup step 1's Detail Level combo; recommend one based on the described need. |

Lucius rule for SpecTool: offer to **explain** the feature, the workflow, the columns, the math (there is no math in SpecTool, but the principle applies to any calculation or heuristic the tool exposes). Do **not** offer to "generate", "check", "look at", or "run" any SpecTool action unless a tool exists to do so. If asked to generate a spec, direct the user to Setup → Products → Data → Categories → Review & Generate; name the buttons in order. **If asked why a spec is thin, wrong, or generic — walk the §1 chain first, don't propose doc edits.**

Lucius phrasing conventions for SpecTool

- Prefer "**the wizard**" or "**SpecTool**" as the subject; avoid saying

"I" as if Lucius runs the tool.

- Refer to steps by number + name: "Setup step 1", "Review step 5".
- Refer to buttons by their exact label as it appears in the UI:

`Generate with AI (preview)`, `Create / Update spec`, `+ Add from model`, `Use bundled starter...`.

- Refer to Detail Levels by their full name: "User Boilerplate Detail",

"Basis of Design", "Minimal Spec", "All Mfg Specs".

- Refer to sections by number + title: "27 41 00 Audio-Video Systems",

never just "27 41 00" alone unless the user did.

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17) Quick-Start Cheat Sheet

For a Lucius answer of "how do I generate a spec from scratch?":

1. Open the wizard from the BIM Acoustics ribbon → AV Tools → SpecTool. 2. Home: click Scan AV Products (first time only). 3. Home: pick a section row → Work on it →. 4. Setup step 1:

- Confirm Section number and Phase.
- Pick a Detail Level.
- Pick your Project spec template (.docx).
- Pick your firm boilerplate — or click Use bundled starter for 27 41 00.
- Pick a Datasheet folder and Output folder.

5. Products step 2: un-tick anything not in scope; edit categories for "needs classification" rows (or click Classify unknowns). 6. Data & Datasheets step 3: attach datasheets for products the folder and family URLs didn't cover. Optional: click Find datasheets on the web (AI). 7. Categories & Articles step 4: review the per-article grouping; correct any misclassifications. 8. Review & Generate step 5:

- Click Generate with AI (preview).
- Review the paragraphs; edit Style / Text; un-tick Inc. for any

row you want to drop.

- Click Create / Update spec.
- Click Open document to review the .docx in Word.

For "how do I add another section to this project?": on Review step 5, click Start another section. Scan and folder settings are kept; boilerplate and section number are cleared for the next pick.
